Autodesk® Scaleform®

Scaleform 3D

Scaleform 3.2 3D

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Autodesk® GAMEWARE

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Autodesk® Scaleform® 4.4

	Scaleform 3D
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1

Scaleform® 3Di™ UI 3D Flash

Scaleform 3.2 ,Scaleform ActionScript , 3D Scaleform 3.2 ActionScript 2.0 , 3D AS --- AS 3.0

(_z _zscale _xrotation _yrotation _matrix3d _perspfov), HUD UI 3D 3D

c + + 3D Scaleform API API 3D GFx::Value

, 3D 3D Flash ,

2 Flash 10

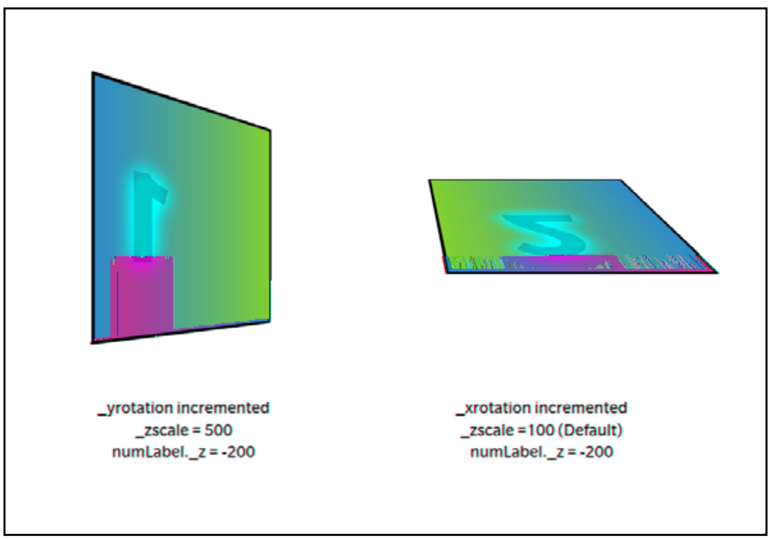
,Scaleform 3Di Flash 10 3D , 3Di
Scaleform AS2

2.1

Flash 10 3D ,3Di :

1. , , ,

2. (Back-face culling)



1 (Culled)

2.2

Scaleform 3Di , Flash 10 3D ,

2D 3D , 3D 3D
Flash 10

3 3D Flash

Scaleform 3.2, 3D Flash

3.1

Flash ,
3D Scaleform 3.2 , 3D Flash

“ ”(Render to Texture) 3Di Flash
3D , 3D , 3D
Z( ) / (UI)

, , , ,
Flash (Scaleform) ,
, Flash ,
, “SWF ”(SWF to Texture) , Scaleform SDK

3.2 3Di

(Scaleform 3.2) 3Di --- ActionScript Direct Access API (C++)
,3Di , ,
– , ,
Scaleform Scaleform

,3Di , Scaleform , UI,
, UI , ,
UI 3D , UI
, UI , ,

4 3Di API

4.1 ActionScript 2 API

3Di,Flash
,

, , 3D , 3D (x y z 3d), , 3D , 3d “ ”(Null)(, ActionScript ,foo._matrix3d = null), 2D z 0 3D

,3D 3 (x y z) , Z 2D ,3D , _zscale ActionScript

Scaleform 3D Flash 10 , +X ,+Y , +Z ( ) +Z ,FOV 55

3D Flash

-
-
-

3Di ActionScript

- **_z** - Z ( ), 0
- **_zscale** - Z , 100
- **_xrotation** - X( ) , 0
- **_yrotation** - Y( ) , 0
- **_matrix3d** - 16 (4 x 4) 3D NULL, 3D , 2D
- **perspfov** – (FOV) > 0 < 180 -1 FOV 55

ActionScript gfxExtensions True( )

, ,
, _xrotation _yrotation 3D ,
, 3Di
,

1

Y 45

```
_global.gfxExtensions = true;
mc1._yrotation = 45;
```

2

X Z

```
_global.gfxExtensions = true;
sq1._zscale = 500;

function rotateAboutXAxis(mc:MovieClip):Void
{
    mc._xrotation = mc._xrotation + 1;
    if (mc._xrotation > 360)
      {
        mc._xrotation = 0;
    }
}

onEnterFrame = function()
{
    rotateAboutXAxis(sq1);
}
```

3

3D 3D

```
function PixelsToTwips(iPixels):Number
```

```
{
      return iPixels*20;
}


function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [ 1, 0, 0, 0,
                0, 1, 0, 0,
                0, 0, 1, 0,
                tX,tY,tZ,1];
    return matrixC;
}

this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);
```

4.2 C++ 3Di

4.2.1 Matrix4x4

Matrix4x4 4x4 3D
( ) , Render_Matrix4x4.h

4.2.2 Direct Access

Direct Access (GFx::Value) 3Di API 3D
3Di ActionScript

, GFx_Player.h GFx::Value::DisplayInfo

```
void    SetZ(Double z)
void    SetXRotation(Double degrees)
void    SetYRotation(Double degrees)
void    SetZScale(Double zscale)
void    SetFOV(Double fov)
void    SetProjectionMatrix3D(const Matrix4F *pmat)
void    SetViewMatrix3D(const Matrix3F *pmat)
bool    SetMatrix3D(const Render::Matrix3F& mat)

Double  GetZ() const
Double  GetXRotation() const
Double  GetYRotation() const
Double  GetZScale() const
Double  GetFOV() const
const Matrix4F* GetProjectionMatrix3D() const
const Matrix3F* GetViewMatrix3D() const
bool    GetMatrix3D(Render::Matrix3F* pmat) const
```

4.2.3 Render::TreeNode

Render::TreeNode , 3D " "(View) "
"(Perspective) , ,

`GFx_Player.h`

```
void      SetProjectionMatrix3D(const Matrix4F& m)
void      SetViewMatrix3D(const Matrix3F& m);
```

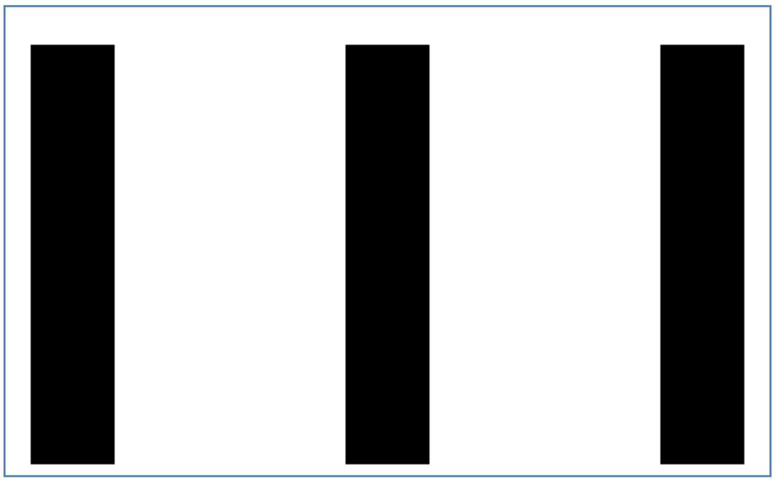
4.2.4 : Direct Access API

```
Ptr<GFx::Movie> pMovie = …;
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
      GFx::Value::DisplayInfo dinfo;
      bOK = tmpVal.GetDisplayInfo(&dinfo);
      if (bOK)
      {   // set perspectiveFOV to 150 degrees
          Double oldPersp = dinfo.GetFOV();
          dinfo.SetFOV(150);
          tmpVal.SetDisplayInfo(dinfo);
      }
}
```

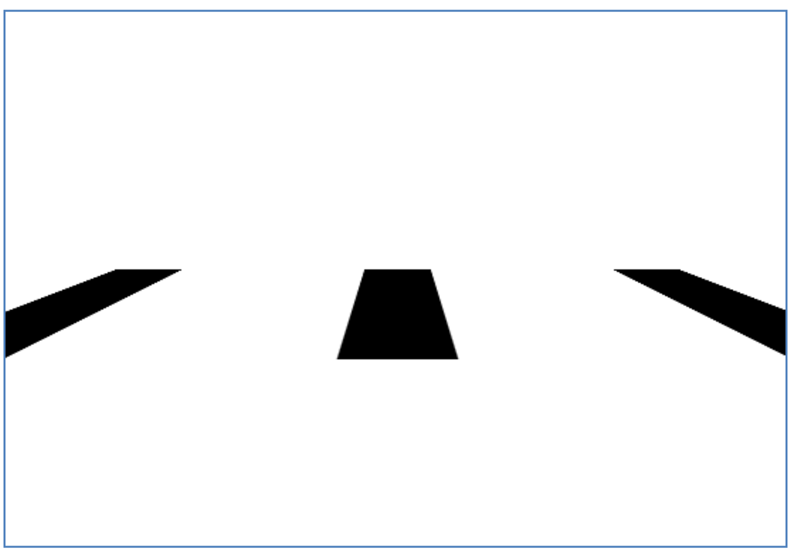
5

Direct Access API (FOV)
FOV 0, FOV

, ,
, z FOV ,
0 180, 1, , 179 ,



2 Flash



3 -80 55 X

FOV (5) . ,

。 ActionScript property _perspfov -1 C++

Direct Access API :

// 1. Perspective FOV -1

```
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
   GFx::Value::DisplayInfo dinfo;
   bOK = tmpVal.GetDisplayInfo(&dinfo);
   if (bOK)
   {
      dinfo.SetPerspFOV(-1);
      tmpVal.SetDisplayInfo(dinfo);
   }
 }
```

// 2.

```
void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)
{
   const float nearZ = 1;
   const float farZ = 100000;
   float DisplayHeight = fabsf(visFrameRectInTwips.Height());
   float DisplayWidth = fabsf(visFrameRectInTwips.Width());
   matPersp->OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root.mc");
if (bOK)
{
   GFx::Value::DisplayInfo dinfo;
   bOK = tmpVal.GetDisplayInfo(&dinfo);
   if (bOK)
   {
      Matrix4F perspMat;
      MakeOrthogProj(PixelsToTwips (pMovie->GetVisibleFrameRect()), &perspMat);
      dinfo.SetProjectionMatrix3D(&perspMat);
      tmpVal.SetDisplayInfo(dinfo);
  }
}
```

(Projection)

```
Ptr<Movie> pMovie = …
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie->GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()),  fabs(visFrameRectInTwips.Height()),
                                                           nearZ, farZ);
pMovie->SetProjectionMatrix3D(ortho);
```

5.1 AS3

ActionScript 3 , PerspectiveProjection( )

Example:

```
import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;
par.y = 100;
par.rotationX = 20;

addChild(par);

var chRed:Sprite = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0, 0, 100, 100);
chRed.x = 50;
chRed.y = 50;
chRed.z = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection = pp3;
```

6 Stereoscopic 3D

Scaleform 3Di UI, 3D
, , , ,
, , ,
,

Scaleform 3Di 3D , ,PC NVIDIA 3D Vision Kit NVIDIA
, , , ( )
, Display , , ,
(Parallax Shift) Scaleform 4 , PS3
TinyPlayerStereo(Scaleform)：
Apps \Samples \GFxPlayerTiny \GFxPlayerTinyPS3GcmStereo.cpp

6.1 NVIDIA 3D Vision

6.1.1 3D Vision

NVIDIA 3D Vision Kit 3Di UI, NVIDIA
3D 3D (Stereoscopic 3D
Test)

6.1.2

Scaleform Player 3D Vision
Direct3D Scaleform Player, , Scaleform Player ,
command line argument –f NVIDIA ,
3D 3D , (CTRL + T)

6.1.3

, ,
, 3D Vision (Convergence)
, ,

, 3D Vision , IR ,
(CTRL + F3) /(CTRL + F4) , , ,
, Scaleform Player ,
(CTRL + F5) (CTRL + F6), , ,

, "Stereoscopic 3D", "Set up stereoscopic 3D", "Set Keyboard Shortcuts" NVIDIA " "(Enable the advanced in-game settings)
, (CTRL + F5) (CTRL + F6) , (CTRL + F7)

, , , ,
(/) , (CTRL + F5), 10 15 , ,
(CTRL + F5) (CTRL + F6), (
) 0 --- ,
, (CTRL + F7) ,

NVIDIA API, 3D , ,
3D ,

API
http://developer.nvidia.com/object/nvapi.html.

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform Stereo API

,Scaleform 4.1 Render::HAL API, 3D 3D
HDMI 1.4 。

, Scaleform , 6.2.1 。

Scaleform 4

Render::Viewport::SetStereoViewport

SF4 (Viewport Stereo)

```
// ; ,

View_Stereo_SplitV        = 0x40,
View_Stereo_SplitH        = 0x80,
View_Stereo_AnySplit    = 0xc0,

void SetStereoViewport(unsigned display);
```

6.2.1 Scaleform Stereo

```
Render::StereoParams s3DInfo;
s3DInfo.DisplayDiagInches = 46;      // 46 inch TV
s3DInfo.DisplayAspectRatio = 16.f / 9.f;
s3DInfo.Distortion = .75;        // 0 to 1
s3DInfo.EyeSeparationCm = 6.4;    // this will default 6.4cm
pRenderHAL->SetStereoParams(s3DInfo);
```

, Display , , Display
Scaleform :

```
pMovie->Advance(delta);

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);
pMovie->Display();
```

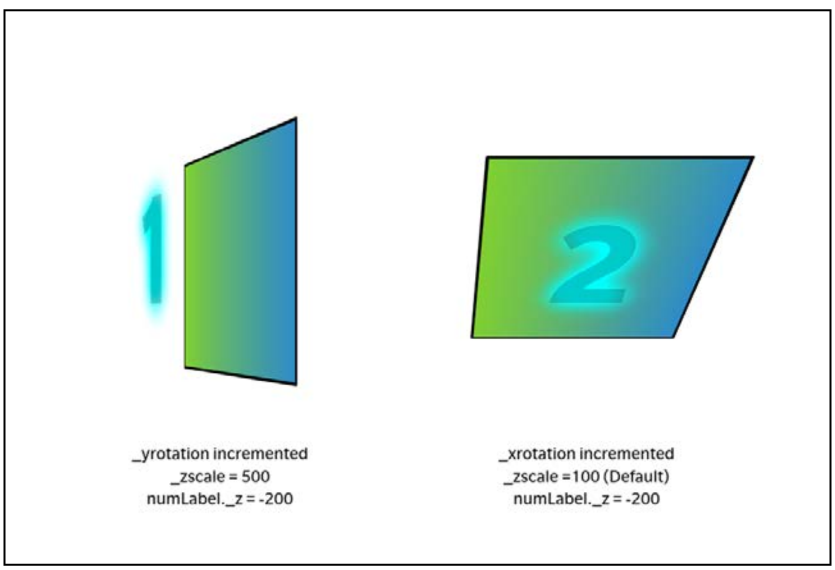
```
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);
pMovie->Display();
```

7

3Di Scaleform SDK 3D Flash (SWF FLA)

, *C:\Program Files (x86)\Scaleform\GFx SDK 4.4\Bin\Data\AS2 or AS3\Samples*

4 3DGenerator



5 3D


3D Flash - tilted using ActionScript, no render textures used
Scaleform GFx
3D Window
Send
Change Image
Enter Text
3D GFx Logo
GFx Logo
Scaleform Logo
Wireframe
© Scaleform
Reset
Quit
Scaleform

6 3D